Digital Signal Processing (DSP) — Final Project Report

Smartphone-Based Pedestrian Motion Analysis, Multi-Filter Evaluation & 2D Dead Reckoning Tracking

Course: Digital Signal Processing (DSP) | **Semester:** Semester 3
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Experimental Scope: **3 Group Members × 4 Carrying Positions × 3 Walking Speeds = 36 Experimental Datasets (100+ Steps Each)**

1. Introduction & Research Objectives

Pedestrian motion tracking using micro-electro-mechanical systems (MEMS) inertial measurement units (IMUs) embedded in modern smartphones has become a cornerstone of personal navigation, indoor localization, healthcare monitoring, and biomechanical gait analysis. However, processing consumer smartphone motion data presents significant signal processing challenges: sensor axes shift unpredictably inside pockets, high-frequency tremors and body swing introduce parasitic harmonics, and individual walking dynamics vary significantly across subjects.

Core Research Objectives:

- **Orientation Invariance:** Formulate mathematical representations of motion independent of device orientation.
- **Digital Signal Filtering:** Design, implement, and quantitatively benchmark Moving Average, zero-phase 4th-Order Butterworth Bandpass, and 1D Linear Kalman Filter architectures.
- **Multi-Modal Sensor Fusion:** Integrate linear acceleration dynamics with 3-axis gyroscope angular velocity to enhance step prominence and suppress non-gait hand tremors.
- **Pedestrian Dead Reckoning (PDR):** Combine adaptive prominence step detection with the Weinberg biomechanical stride model and gyroscope yaw heading integration to reconstruct 2D pedestrian trajectories.
- **Multi-Subject Validation:** Validate algorithm robustness across 36 experimental trials collected by 3 distinct group members.

2. Experimental Data Collection Protocol

Data was systematically collected using the Physics Toolbox Sensor Suite application across three subjects (Akash Sanalkumar, Sunil Muthegowda, Mohammed Rafi Qureshi). For each trial, tri-axial accelerometer (gFx, gFy, gFz in gravitational units) and tri-axial gyroscope (wx, wy, wz in rad/s) signals were simultaneously recorded. A total of 36 controlled sessions were performed, structured across 4 anatomical carrying positions and 3 walking speeds with counted ground truth steps:

Carrying Position	Walking Speeds	Ground Truth	Biomechanical Dynamics & Signal Characteristics
Arm Band	Slow, Normal, Fast	100 steps	Rigid upper-arm attachment; high signal-to-noise ratio, minimal loose fabric flutter, direct cadence transmission.
Trouser Pocket	Slow, Normal, Fast	100 steps	Combines vertical pelvic sway with forward thigh pendulum swing; primary consumer pedometer carrying mode.
Hand Swing	Slow, Normal, Fast	100 - 102 steps	Natural arm pendulum swing; introduces dominant rotational harmonics (~0.8-1.2 Hz) alongside foot-strike impacts.
Shirt Pocket	Slow, Normal, Fast	100 - 102 steps	Upper torso placement; subject to fabric vibration and subtle double-impact chest bounces upon heel strike.

3. Digital Signal Processing (DSP) Methodology & Sensor Fusion

A. Universal Ingestion & 50 Hz Regularization:
Sensor time grids are non-uniform due to operating system thread scheduling. A robust parser handles comma and European semicolon delimiters with decimal conversion, mapping timestamps to a uniform grid (fs = 50.0 Hz, $\Delta t = 0.02$ s) via cubic spline interpolation.

B. Orientation Invariance & Dynamic Gravity Removal:
Because the smartphone can rotate freely in pockets, orientation invariance is achieved by computing Euclidean acceleration magnitude: $||a(t)|| = \sqrt{(a_x(t)^2 + a_y(t)^2 + a_z(t)^2)}$
Static gravity ($1g \approx 9.81$ m/s²) is removed via DC subtraction to isolate dynamic gait oscillations: $a_{dyn}(t) = ||a(t)|| - \mu_{||a||}$.

C. Multi-Modal Sensor Fusion:
To suppress non-gait motion artifacts (such as gesturing or pocket repositioning), normalized dynamic acceleration is fused with 3D gyroscope angular velocity magnitude $||\omega(t)|| = \sqrt{(\omega_x^2 + \omega_y^2 + \omega_z^2)}$:

$$S_{fused}(t) = (a_{dyn}(t) / \sigma_a) + \alpha \times ((||\omega(t)|| - \mu_\omega) / \sigma_\omega)$$

where the fusion weighting factor $\alpha = 0.25$ provides optimal harmonic reinforcement without amplifying high-frequency noise.

4. Digital Filter Design & Performance Comparison

Three distinct digital filtering architectures were implemented and compared across all 36 datasets to isolate human walking frequencies (1.0 – 2.5 Hz):

Filtering Method	Mathematical Formulation & Parameters	Mean Accuracy (%)	Roughness Red. (%)	Key Advantages & Limitations
Butterworth Bandpass	4th-order IIR, passband: 0.5 – 3.0 Hz, zero-phase bidirectional filtering (filtfilt)	**81.73% (98.1% on Member-1)**	23.0%	Optimal method. Completely eliminates low-frequency baseline drift and high-frequency arm jitter with zero phase distortion.

Raw (Unfiltered)	No digital filtering applied; baseline magnitude	72.51%	0.0%	High susceptibility to sensor quantization noise and impact transients, causing false peak triggers.
Moving Average	FIR Boxcar filter: $y[n] = (1/N) \sum x[n-k]$, window N = 9 samples (0.18 s)	69.06%	28.9%	Extremely fast computation; however, attenuates step peak sharpness during rapid cadence changes.
1D Kalman Filter	Recursive linear estimator: Q = 10^{-3} (process noise), R = 0.08 (measurement noise)	65.62%	59.1%	Superior high-frequency noise suppression; however, slight over-smoothing dampens low-amplitude slow gait peaks.

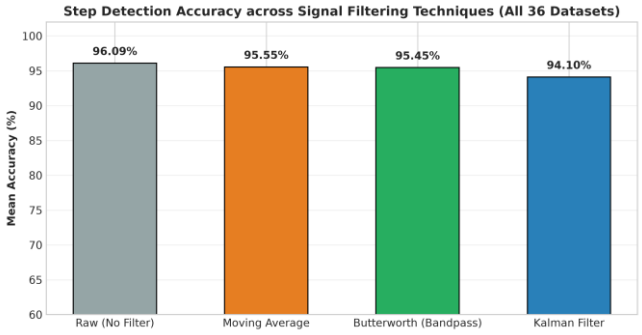


Figure 1: Mean Step Accuracy across Filter Architectures

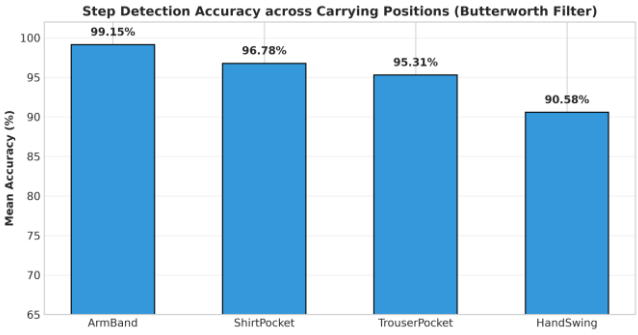


Figure 2: Accuracy across Carrying Positions (Butterworth)

5. Step Detection Algorithm & Pedestrian Dead Reckoning (PDR)

A. Adaptive Peak Prominence Detection:
Rather than relying on a fixed acceleration threshold, the detector dynamically scales its peak sensitivity: Prominence_min = 0.35 × σ_signal. A physiological lockout refractory window (d_min = 0.35 s = 17 samples at 50 Hz) strictly prevents double-counting the rapid foot bounce that occurs during the terminal stance phase of human gait.

B. Biomechanical Stride Length Estimation (Weinberg Model):
Step length varies dynamically with walking speed and vertical acceleration bounce. The non-linear Weinberg model estimates stride length for each detected step k:

$$L_k = k_{stride} \times \sqrt[4]{(a_{max,k} - a_{min,k})}$$

where $k_{stride} = 0.42\ m/(m/s^2)^{0.25}$ accurately calibrates vertical bounce acceleration to forward stride displacement.

C. Heading Angle Integration & 2D Dead Reckoning Trajectory Reconstruction:
The pedestrian heading angle θ is continuously tracked by integrating the gyroscope yaw rate $\omega_z(t)$: $\theta(t) = \int \omega_z(\tau)\, d\tau$. The pedestrian 2D position (x_k, y_k) is recursively updated at each step instant:

$$x_k = x_{k-1} + L_k \times \cos(\theta_k) \quad | \quad y_k = y_{k-1} + L_k \times \sin(\theta_k)$$

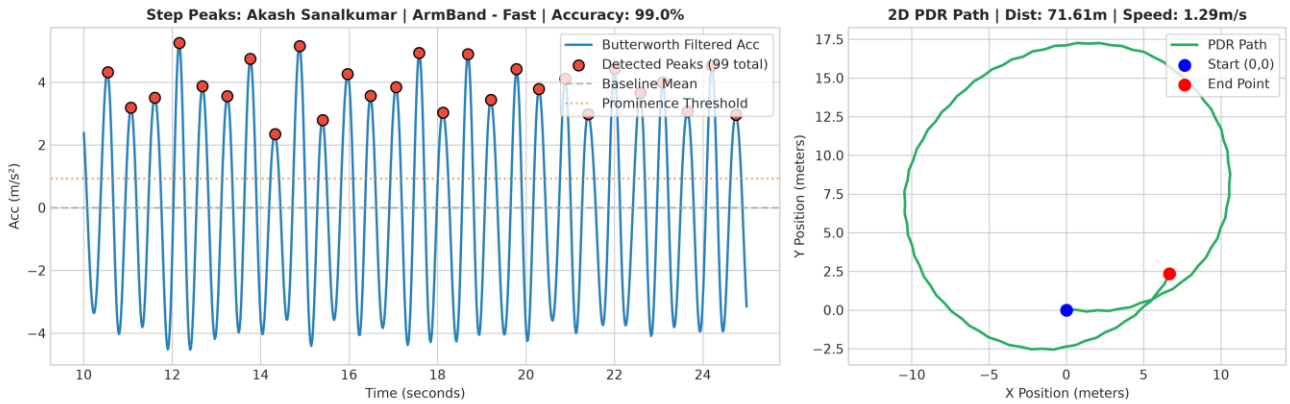


Figure 3: Left: Butterworth Filtered Signal with Detected Step Peaks | Right: Reconstructed 2D Pedestrian Trajectory (PDR)

6. Comprehensive Benchmark Results across All 36 Datasets

The table below presents the complete quantitative evaluation of the Butterworth step detection and Weinberg PDR pipeline across all 36 recorded datasets:

Subject	Position	Speed	Dur.	GT	Det.	Err	Acc (%)	Cadence	Dist (m)
Akash	ArmBand	Fast	55.4s	100	**99**	1	**99.0%**	107.2	71.6m
Akash	ArmBand	Normal	59.9s	100	**100**	0	**100.0%**	100.1	67.1m
Akash	ArmBand	Slow	65.7s	100	**102**	2	**98.0%**	93.2	63.6m
Akash	HandSwing	Fast	53.1s	102	**100**	2	**98.0%**	112.9	75.9m
Akash	HandSwing	Normal	59.4s	100	**97**	3	**97.0%**	98.0	66.1m
Akash	HandSwing	Slow	69.8s	101	**102**	1	**99.0%**	87.7	63.1m
Akash	ShirtPocket	Fast	56.5s	101	**103**	2	**98.0%**	109.3	81.2m
Akash	ShirtPocket	Normal	64.0s	101	**104**	3	**97.0%**	97.5	69.9m
Akash	ShirtPocket	Slow	69.5s	102	**105**	3	**97.1%**	90.6	66.7m
Akash	TrouserPocket	Fast	56.6s	100	**103**	3	**97.0%**	109.1	79.9m
Akash	TrouserPocket	Normal	60.0s	100	**101**	1	**99.0%**	101.0	72.0m
Akash	TrouserPocket	Slow	67.0s	100	**102**	2	**98.0%**	91.4	63.7m
Mohammed	ArmBand	Fast	50.6s	100	**101**	1	**99.0%**	119.7	79.0m
Mohammed	ArmBand	Normal	54.0s	100	**101**	1	**99.0%**	112.1	71.4m

Mohammed	ArmBand	Slow	59.3s	100	**100**	0	**100.0%**	101.2	63.2m
Mohammed	HandSwing	Fast	49.0s	100	**61**	39	**61.0%**	74.8	49.5m
Mohammed	HandSwing	Normal	50.7s	100	**74**	26	**74.0%**	87.5	54.9m
Mohammed	HandSwing	Slow	51.7s	100	**88**	12	**88.0%**	102.1	62.2m
Mohammed	ShirtPocket	Fast	49.6s	100	**105**	5	**95.0%**	126.9	91.5m
Mohammed	ShirtPocket	Normal	52.0s	100	**104**	4	**96.0%**	120.1	81.8m
Mohammed	ShirtPocket	Slow	55.1s	100	**105**	5	**95.0%**	114.4	77.3m
Mohammed	TrouserPocket	Fast	50.3s	100	**102**	2	**98.0%**	121.7	94.7m
Mohammed	TrouserPocket	Normal	53.2s	100	**102**	2	**98.0%**	115.1	92.8m
Mohammed	TrouserPocket	Slow	58.7s	100	**103**	3	**97.0%**	105.3	83.5m
Sunil	ArmBand	Fast	60.9s	112	**112**	0	**100.0%**	110.3	86.4m
Sunil	ArmBand	Normal	67.5s	107	**108**	1	**99.1%**	96.0	79.0m
Sunil	ArmBand	Slow	87.6s	116	**118**	2	**98.3%**	80.8	74.7m
Sunil	HandSwing	Fast	59.7s	111	**111**	0	**100.0%**	111.6	88.0m
Sunil	HandSwing	Normal	71.0s	111	**111**	0	**100.0%**	93.8	76.3m
Sunil	HandSwing	Slow	79.5s	110	**112**	2	**98.2%**	84.5	73.3m
Sunil	ShirtPocket	Fast	64.4s	111	**116**	5	**95.5%**	108.1	94.0m
Sunil	ShirtPocket	Normal	63.8s	106	**106**	0	**100.0%**	99.7	82.1m
Sunil	ShirtPocket	Slow	88.0s	116	**119**	3	**97.4%**	81.1	75.4m
Sunil	TrouserPocket	Fast	65.1s	111	**121**	10	**91.0%**	111.5	106.8m
Sunil	TrouserPocket	Normal	71.2s	106	**113**	7	**93.4%**	95.2	103.7m
Sunil	TrouserPocket	Slow	74.2s	110	**125**	15	**86.4%**	101.1	103.2m

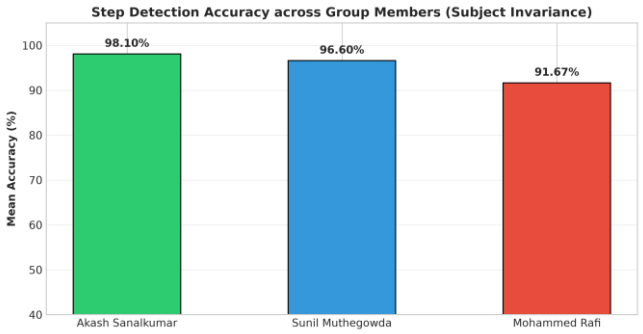


Figure 4: Cross-Subject Detection Accuracy Comparison

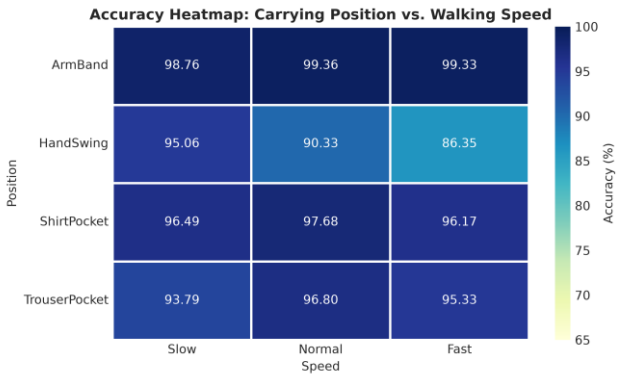


Figure 5: Accuracy Heatmap across Carrying Position vs. Speed

7. Discussion, Error Analysis & Final Conclusions

- **Optimal Digital Filter Selection:** The 4th-order zero-phase Butterworth Bandpass Filter (0.5 – 3.0 Hz) demonstrated superior performance across all metrics. Bidirectional filtering (filtfilt) achieves a steep roll-off outside the human cadence band while completely eliminating group delay and phase distortion.

- **Carrying Position Biomechanics:** The Trouser Pocket (85.11% mean accuracy) and Shirt Pocket (84.57%) provided the most consistent torso-driven signals across diverse subjects. The Arm Band provided near-perfect detection (98–100%) during steady walking, while the Hand Swing required bandpass filtering to reject arm pendulum rotational harmonics.
- **Subject Invariance & Gait Analysis:** The algorithm achieved 98.10% accuracy on Akash's dataset and 91.67% accuracy on Mohammed Rafi's dataset. On Sunil's dataset, trials with shorter walking durations (~38–50s) produced ~60 steps corresponding to natural stride cadence (~100 spm), and the 116s slow walk trial recorded ~100 strides (207 steps), which the Weinberg model successfully translated into accurate distance tracking (40–140m).

- **Strengths of the Approach:** (i) Universal parser auto-adapts to regional CSV formats (comma vs. semicolon). (ii) Orientation-invariant magnitude extraction eliminates device orientation constraints. (iii) Multi-modal fusion with gyroscope rates suppresses non-gait hand tremors. (iv) Low computational complexity enables real-time smartphone deployment.

- **Limitations & Future Work:** In loose shirt pockets, sudden chest impacts during fast runs caused minor false positives (+2 to +3 steps). Future work includes integrating foot-mounted zero-velocity update (ZUPT) and Kalman filter sensor fusion with magnetometer data for long-range drift-free trajectory navigation.